Problem Set 2. Probability
Quantitative Methods for Humanities

1. **Newspaper Headlines and Event Unions.**

   A news article (hypothetical) shows that out of 500 front-page articles in a major newspaper:

   - 200 are related to politics (Event P).
   - 150 are related to social issues (Event S).
   - 60 are related to both politics and social issues.

   Answer the following questions

   (a) What is the probability that a randomly chosen front-page article is about politics or social issues?

   (b) If you know an article is about politics, what is the probability it also involves social issues?

2. **Polling Philosophers: Independence or Not?**

   A philosophical society publishes a study on 300 philosophers. For a given philosopher, define the events:

   - R: "The philosopher self-identifies as a Rationalist."
   - E: "The philosopher primarily studies Ethics."

   It is known that:

   $$P(R) = 0.4, \quad P(E) = 0.25, \quad P(R \cap E) = 0.12.$$

   (a) Check if R and E are independent by comparing $P(R \cap E)$ with $P(R) \times P(E)$.

   (b) Interpret the result.

3. **Conditional Probability in Political Prediction.**

   Consider the following political prediction problem focused on forecasting the outcome of a presidential election:

   (i) Historically, there is a 1 in 4 chance that a candidate from Party A will win the presidency in any given election.

   (ii) A political polling system categorizes the likelihood of Party A's win as high probability based on campaign trends and voter behavior.

(iii) Out of 100 elections won by Party A, the polling system identifies high probability in 90 cases.

(iv) There is a 1 in 5 chance that Party A will not win the election, yet the polling system still predicts high probability.

Answer the following questions:

(a) Define the probabilities $P(\text{Win})$, $P(\text{HP} \mid \text{Win})$, $P(\text{HP} \mid \text{Lose})$, and $P(\text{Lose})$.

(b) Using Bayes' rule, compute $P(\text{Win} \mid \text{HP})$.

(c) Interpret the result in the context of the polling system's prediction.

4. **Bayes' Rule for Fake News Detection.**

The Collins Dictionary named "fake news" the 2017 term of the year. And for good reason. Fake, misleading, and biased news has proliferated along with online news and social media platforms which allow users to post articles with little quality control.

In this problem we will implement a basic news "reality detector" using Bayes Rule. To this end, we'll examine a sample of 150 articles which were posted on Facebook and fact checked by five BuzzFeed journalists[1]. Information about each article is stored in the `fake_news` dataset in the `bayesrules` package. To learn more about this dataset, type `?fake_news` in your console.

(a) Display a summary of the data.

(b) Summarize in a table the total and relative numbers of fake (event F) and real articles (variable `type`)

(c) We will say that an article "appears sensational" (event S) if its title contains and exclamation mark (`title_excl` variable). How many articles "appear sensational" among fake and real ones?

(d) What is the probability $P(S \mid F)$

(e) Use Baye's rule to compute compute the probability of an article being fake given that it "appears sensational"

Discussion: This approach illustrates how evidence (an article appearing sensational) can update our belief about its authenticity. Similar methods can be incorporated into fact-checking algorithms that continuously update the probability of a news article being fake as more features or evidence are observed.

The take away is that we most take advantage of upcoming or extra information available to **update our prior believes of how likely is the occurrence of an event**, and Bayes' rule provide us a method to do this.

5. **Bayesian Analysis of Stylometric Testing.**

It has been suggested[2] that an anonymous novel published in 2010 might have been written by the American novelist Thomas Pynchon. A stylometric test is used to determine the likelihood of Pynchon's authorship. The test has the following properties:

[1] Shu, Kai, Amy Sliva, Suhang Wang, Jiliang Tang, and Huan Liu. 2017. "Fake News Detection on Social Media: A Data Mining Perspective" ACM SIGKDD Explorations Newsletter 19 (1): 22–36.

[2] Art Winslow. The Fiction Atop the Fiction. September 2015. link.

(a) The probability that a random work of literary fiction published between 1960 and 2010 was written by Pynchon (or under his pseudonym) is 1 in 100,000.

(b) The test correctly identifies a work by Pynchon 90% of the time.

(c) The test incorrectly attributes a work to Pynchon 1% of the time.

The novel in question, published in 2010, tested **positive** for Pynchon's authorship. What is the probability that the novel was actually written by Pynchon?

6. **The Enigma machine.**

During World War II, five cryptanalytic machines were developed to decode 1500 messages encrypted by the Enigma cipher machine. The Table 1 below presents information on the number of messages assigned to each machine and the machine's failure rate (i.e., the percentage of messages the machine was unable to decode). Aside from this information, we do not know anything about the assignment of each message to a machine or whether the machine was able to correctly decode the message.

Machine	**Number of Messages**	**Failure Rate**
Banburismus	300	10%
Bombe	400	5%
Herivel tip	250	15%
Crib	340	17%
Hut 6	210	20%

Table 1: Number of messages and failure rates of cryptanalytic machines

Suppose that we select one message at random from the pool of all 1500 messages but find out this message was not properly decoded. Which machine was most likely responsible for this mistake?

7. **Passing a Quiz by Randomly Answering.**

You have a quiz tomorrow on the subject "Quantitative Methods for Humanities" and have not studied at all, hopping that the chances of passing (at least 5 points out of 10) by randomly answering are not too low.

The quiz consists of 10 multiple-choice questions. Each question has 4 possible answers, and only two are correct (this information is not known to you). A question is correctly answered only if both correct options are selected as true. Each correctly answered question earns 1 point.

a) Using counting techniques, what is the probability of correctly answering a question?

b) By modeling the number of correctly answered questions as a binomial random variable, was is the probability that you passes the quiz?

c) Repeat the calculations of the previous parts by assuming that now you know that only two are of the four choices are correct for each question.

8. **The pardoned prisoners.**

There are three prisoners: A, B, C. One prisoner, selected uniformly at random, will be pardoned while the other two will be executed. Prisoner A asks the jailer (who knows) to tell him one of the

prisoners who will be executed: "If I will be pardoned, just flip a coin and tell me B or C; if C will be pardoned, tell me B; and if B will be pardoned, tell me C." The jailer says B will be executed.

(a) Using Bayes Rule, what is the numerator of the expression evaluating the probability that C is pardoned?

  (i) P(B is executed | C is pardoned) P(B is executed); or

  (ii) P(jailer says B executed | C is pardoned) P(C is pardoned); or

  (iii) P(C is pardoned | jailer says B is executed) P(B is executed)?

(b) What is the probability of C being pardoned?

9. **Guessing the fair coin.**

You have a bucket with 2 coins: one is a regular coin, while the other has heads on both sides. You pick a coin at random and flip it: it turns up heads. What is the probability you chose the fair coin?

10. **Making fair the unfair coin.**

An unfair coin lands head with probability $p > 0.5$. Cleverly, you designed the following method to make the coin fair: toss the coin twice, if the same side appear in both trials, start again, if two different sides appear, take the last tossing as the valid one. Prove that this new method makes the probability of each side equal. What would happen if the coin is tricked in the sense that has heads in both sides?

11. **Testing Psychic Abilities with the Zener Test.**

The **Zener Test** is an experiment used to test for extrasensory perception (ESP). A researcher shows a subject a deck of 25 cards, each with one of five symbols (circle, cross, waves, square, star). The subject must guess the symbol on each card before seeing it. (Curious about your result? Take the test here)

(a) Compute the probability of randomly guessing the correct symbol.

(b) Given that the subject completes a 25-card Zener Test, model the number of correct guesses X using a random variable.

(c) What is the expected number of correct guesses if the subject has no psychic ability?

(d) What is the probability of getting exactly 10 correct guesses?

(e) What is the probability of getting 15 or more correct guesses?

(f) If a subject correctly guesses 15 or more cards, would this be strong evidence for psychic ability? Why or why not?

(g) If 1,000 people take the test, what is the probability that at least one person gets 15 or more correct guesses?

(h) If we increase the number of test participants, this probability also increases. Does that mean we can prove the existence of psychic abilities simply by increasing the sample size? Discuss this seemingly "paradox".

12. **Mean and variance.**

    Suppose I give you a random variable X, such that its mean is 0 and that the expectation of X^2 is 5. What is the variance of X?

13. **Anthropology study of weight.**

    It is known that the elderly women of a certain rural American population have a mean weight of $\mu = 150$ pounds, with a standard deviation of $\sigma = 10$. Assuming the weight follows a normal distribution, find:

    (a) The probability of selecting a woman who weighs between 125 and 135 pounds.

    (b) The probability of selecting a woman who weighs between 140 and 160 pounds.

    Now drop the normality assumption and find:

    (c) An approximation of the probability that a sample of size $n = 30$ has a sample mean $\bar{Y} = 153$. What would your answer be if $n = 100$?

    What can you conclude from the results?

14. **Beating the Casino?**

    A casino offers a simple game: A player bets €1, and with probability 0.49, they win €2. Otherwise, they lose their €1 bet.

    (a) Compute the expected net profit per bet.

    (b) Suppose a gambler plays 10 times. Could they walk away with a profit? With what probability?

    (c) What if she plays $10,000$ times? Use the Central Limit Theorem. Is this method accurate for the previous part where the gambler payed only 10 times?

    (d) Use the Law of Large Numbers to give an approximation of the profit the casino makes after n bets, for large n. Use this estimation to explain how the casino ensures that, over a long period, it will always make a profit, even though some individuals may win in the short run.

    (e) A player has lost 5 times in a row and believes they are now "due for a win". Explain why this reasoning is incorrect.

15. **Democracy and the Condorcet's Jury Theorem (CJT)[3].**

    The CJT states that, when certain assumptions hold, the probability that a majority of voters support the correct decision increases and approaches one as the number of voters increases. The assumptions are:

    (i) each voter is more likely than not to identify the correct decision (the competence assumption);

    (ii) voters vote for what they believe is the correct decision (the sincerity assumption);

[3] Condorcet, Marquis de, 1785, Essai sur l'application de l'analyse à la probabilité des décisions rendues àla pluralité des voix, Paris; reprinted Cambridge: Cambridge University Press, 2014. DOI: 10.1017/CBO9781139923972

(iii) votes are statistically independent of one another (the independence assumption).

Introduce the notation

$$\begin{aligned} p &= \text{probability that an individual voter makes the correct decision;} \\ N &= \text{total number of voters;} \\ X &= \text{number of voters who vote correctly.} \end{aligned}$$

Assume that N is off to avoid the posibility of a tie. Prove the theorem by following this steps:

(a) What is the distribution of X? What are its mean and variance?

(b) Express the probability of the right choice winning using X's distribution.

(c) Compute the probability of choosing the correct decision for $N = 5, 15, 25$ and $p = 0.6$. Comment.

(d) Use the CLT to approximate the distribution of X for large N.

(e) Use the CLT to express the probability of the right choice winning in terms of a standard normal random variable.

(f) Take $N \to \infty$ to conclude the proof.